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Title: Sedation and Anesthesia protocol for patients with mitochondrial and fatty acid oxidation disorders

1. Introduction:

It is increasingly common for children with mitochondrial disease to undergo surgery and anesthesia. Although many different anesthetics have been used successfully for these patients, serious, unexpected complications have occurred during and following anesthetic exposure. Defects in function of the mitochondrial electron transport chain can lead to striking hypersensitivity to anesthetics in children. They may face an increased risk of perioperative complications such as respiratory depression, worsened cardiac function and arrhythmias, metabolic disturbances, or severe neurologic injury. Respiratory depression may occur simply from the combination of anesthetic drugs and pre existing muscle weakness seen in any myopathic state. Part from this, a risk of malignant hyperthermia like manifestations or more commonly perioperative rhabdomyolysis is a possibility in patients with mitochondrial diseases. These patients are at risk of metabolic decompensation, including lactic acidosis and encephalopathy with preoperative fasting, perioperative stress, and pain

Organ systems with high metabolic requirements are uniquely dependent on the energy delivered by mitochondria. Thus mitochondrial dysfunction most commonly affects the function of the central nervous system, the heart, the gastrointestinal tract and the muscular system. Most of the medications used for general anesthesia are depressants of one or more of these systems; great care must be exercised in managing their use. Mitochondrial patients often require **smaller doses** of general anesthetics, local anesthetics, sedatives, analgesics, and neuromuscular blockers (paralytics) to achieve the desired end points. Furthermore, it is important to avoid increasing the metabolic burden of patients with MD by not requiring prolonged fasting, and preventing hypoglycemia, postoperative nausea and vomiting, hypothermia (with resulting shivering), prolonged orthopedic tourniquet application, acidosis, and hypovolemia.

2. Recommended:

2.1 Pre operative evaluation: careful evaluation to establish the baseline comorbidities and degree of organ dysfunction especially cardiomyopathy, arrhythmias, respiratory weakness or insufficiency, obstructive sleep apnea, seizures and lactic acidosis.

2.2 Delay elective surgery if there is any evidence of infection

2.3 Pre procedure fasting:Avoid prolonged fasting. Start maintenance / 1.5 maintenancefluids when NPO

Clear fluids	2 hours before the procedure

Infant Formula	6 hours before the procedure
Solids and Nonhuman Milk	6 hours before the procedure

2.4 Intravascular volume and hydration status require close attention, with no administration of lactate containingfluid.

2. 5 Addition of intravenous glucose of minimal of Dx10% dextrose.Age and weight adapted glucose infusion (aiming 6-9 mg/kg/min)

2.6 If the patient is on ketogenic diets for control of seizure, don't administer glucose fluids

2.7 Continue the ongoing treatment e.g vitamins and co enzymes

2.8. Preand post procedure glucose, CK monitoring along with blood gas and lactate. Don't attempt any processure of patient hypoglycemic.

2.9 Monitoring:

- Standard ASA monitoring: ECG, HR, SpO2, NIBP, and end tidal CO2 (ETCO2)
- Continuous temperaturemonitoring
- Bispectral Index (BIS) or other similar systems if available
- Blood sugar
- Neuromuscular blockade

2.10 Use intravenous fluid warmers and active patient warming and cooling devices to maintain euothermia

2.11 Aim for opioid sparing techniques

2.12 Airway management: Mask, laryngeal mask airway, or endotracheal intubation may be chosen depending on the patient's respiratory status and risk for aspiration.

2.13 Careful padding of pressure points and patient positioning are essential, as a subclinical neuropathy may be present.

2.14 Application of surgical tourniquets should be minimized

3. Medication

Opioids	Remifentanil infusion, Fentanyl boluses or infusion can be used
	Morphine to be avoided
Ketamine	Analgesic blouse doses can be used
Dexmedetomidine	Continuous infusion can be used, but clearance may be slowed and effect prolonged
Barbiturates	To be avoided
Midazolam	Use with caution. Avoid in presence of catabolic stress

Propofol	To be avoided
Volatile anesthetics	Not contraindicated. Although some patients may be more sensitive to them. If used should carefully monitor the depth of anesthesia and the anesthetic dose
Muscle relaxant	Try to avoid. If required, nondepolarizing muscle relaxants are preferrede.g rocuronium, cisatracurium, mivacurium
	Depolarizing muscle relaxant succinylcholine: to be avoided
Local anesthetics	Lidocaine and Ropivacaine are preferred
	Bupivacaineto be avoided
NSAIDs	Can be used
Acetaminophen	Avoid repeated doses
Try to avoid the following medication: Mitochondrion-toxic agents include corticosteroids, valproic acid, statins, fibrates, biguanides, glitazones, beta-blockers, amiodarone	

4. Parental nutrition:

Intralipid is contraindicated in patient with fatty acid oxidation deficiency and carnitine uptake defect disorders.

Avoid keeping the patient for more than 48 hrs with zero protein intake which could eventually lead to catabolic state and metabolic crisis.

References:

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2-Perioperative measures in very long chain acyl-CoA dehydrogenase deficiency

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3. A comparison of the effects of midazolam, propofol and dexmedetomidine on the antioxidant system: A randomized triaAuthors:Chao Han, Weiliang Ding, Wenjie Jiang, Yu Chen, Dongyuan Hang, Daming Gu, Guojun JianYongfei Tan, Zhijun Ge , TieliangMa

4-Increased Sensitivity to Rocuronium and Atracurium in Mitochondrial Myopathy

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Back